

Unit-Nets: One Day of Training Without Gradients in a Constrained Weight Geometry

Remy & Claude

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1 The task

Train an MNIST classifier under hard structural constraints (a *unit-net*): every layer carries two nonnegative matrices, excitatory P and inhibitory N , with pre-activation $z = Pa - Na$ (dual projections: one input may excite *and* inhibit the same neuron); each row of P sums to exactly 1 (convex combination); $N \in [0, 1]$ per synapse; all activations provably in $[0, 1]$ (inputs in $[0, 1]$ + convex rows force $z \leq 1$); a constant-1 feature appended per layer is the only bias. Architecture: $784(+1) \rightarrow 64(+1) \rightarrow 10$, ReLU, 101,780 parameters. The driving question: can simulated annealing with beam search — moves of $\pm\delta$ on an edge, parallel-row perturbation with renormalization, and moving δ between two edges into the same neuron — replace backprop here, and at what resource cost?

2 Baselines and the geometry’s pathologies

The oracle is backprop through a constraint-satisfying reparameterization ($P = \text{rowsoftmax}(W)$, $N = \sigma(V)$). Four traps, each a direct consequence of the geometry: (1) $Pa \leq 1$ but Na can reach n_{in} — N must initialize near zero or every ReLU is dead; (2) weight decay on V pulls $\sigma(V)$ toward 0.5, *growing* inhibition — regularize N itself (L1), never its reparameterization; (3) convex rows are averaging operators, so cross-entropy needs logit scale 60–100 to see any spread; (4) a row of N can sum to hundreds, enabling margin-memorization that collapses test accuracy ($97 \rightarrow 80\%$) — validation selection is mandatory. Tuned: **97.2%** (H=64) / **97.7%** (H=256) in ~ 5 s. A fairly tuned *unconstrained* twin reaches 97.7% (H=64) / 98.2% (parameter-matched H=128): the constraints cost ~ 0.5 –1 point of ceiling but give the smallest generalization gap on the board (0.2 vs 0.5).

3 Annealing with beam search

Beam of 8; 31 mutants per member per step; 256 candidates scored on a shared minibatch as one batched matmul; Boltzmann selection at annealed temperature. The move vocabulary (all constraint-preserving by construction):

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- A one inhibitory weight $\pm\delta$, clamped to $[0, 1]$
 - B perturb a neuron’s whole positive row, renormalize
 - C move δ between two positive weights of one neuron (sum conserved)
 - D Gaussian-blob mass tilt on a first-layer row (radius annealed $6 \rightarrow 0.8$ px)
 - E heat-kernel blob on the activation-correlation graph (Hebbian topology)
 - F blob over an imposed 8×8 hidden lattice
 - G blob-pair transfer: take under blob 1, deposit as blob 2 ($\rightarrow$ C as radius $\rightarrow 0$)
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Findings: **selection noise, not ruggedness, caused every plateau** (growing the scoring batch $256 \rightarrow 2048$ turned a hard 76% plateau into monotone climbing); **move design dominates schedule tuning** (A+C

matched A+B+C at 1/6 the compute; whole-row noise B is dead weight); **structured proposals beat temperature** (blob move: +3.2 points free — convolution’s locality injected through the proposal distribution, no weight sharing); **topology beats geometry** (imposed 8×8 lattice 90.4%; topology derived from the network’s own activity via e^{-tL} on the correlation graph: **91.4%**, the gradient-free record — Hebb applied to the search operator); **coarse-to-fine is universal** (radii, edit counts, temperature, and the move mix itself all want to anneal together).

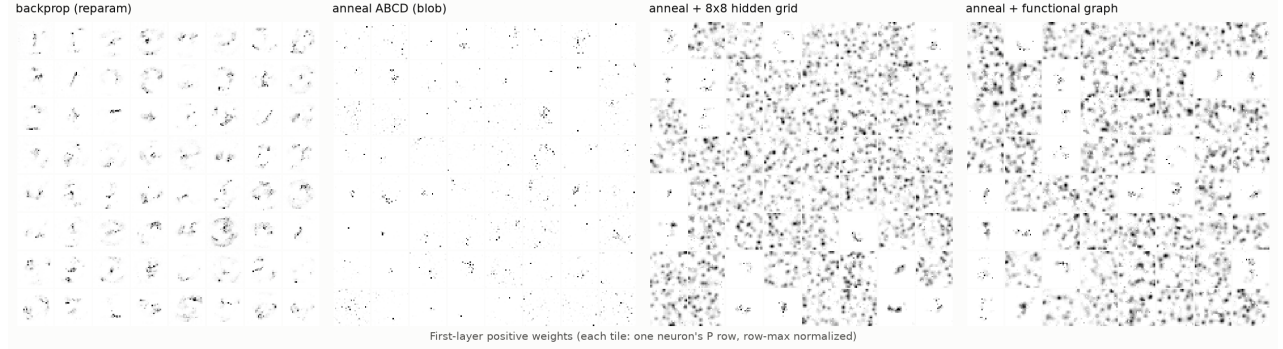


Figure 1: First-layer P -rows. Backprop learns sparse stroke templates; plain annealing leaves salt-and-pepper; topology-aware moves grow dense, contiguous receptive fields.

4 Event-driven scoring (delta evaluation)

A mutation touches one row, so scoring needs no full forward: cache parent activations, propagate the rank-1 consequence ($dz_r = a \cdot \Delta W_{\text{row}}$, one outer product through the output column). Exact (self-checked), $\sim 50\times$ fewer FLOPs; mutants become 3KB edit descriptors instead of 400KB weight copies. The annealing-vs-backprop gap fell from $\sim 6,700\times$ to $\sim 140\times$.

5 The gradient bridge

“The gradient of the transfer” is classical: per-row optimal transfer is pairwise Frank–Wolfe; applying it to all coordinates multiplicatively is exponentiated gradient / **mirror descent** (row $\leftarrow$ row $\cdot e^{-\eta g}$, renormalize) — native to the simplex, no reparameterization, sparse boundary reachable without softmax saturation. The information ladder (matched budgets):

Update rule	gradient information per row	test
random proposals + selection (lattice annealer)	none	90.8%
top- k pairwise Frank–Wolfe ($k=32$)	64 coords, uniform	92.4%
signSGD, fixed δ (Remy’s hybrid)	1 bit/weight	92.7%
mirror descent	full magnitudes	95.0%
Adam + reparameterization	magnitudes + curvature memory	97.2%

Two null results with content: heat-kernel *smoothing* of gradients is worth $\approx$ nothing — the spatial prior worth +3.2 points to blind search is redundant once gradients exist; and a REINFORCE proposal policy **learned Hebb’s rule from reward alone** (sink $\propto$ pre-post correlation, source anti-correlated) yet slightly hurt accuracy: greedy proposals collapse the diversity selection feeds on.

6 Planners: the row solver and MCTS

The output layer is jointly convex given hidden features. Solving it exactly: finished annealed models gained only +0.3–0.4% (readouts already block-optimal); random convex features + solved readout = *chance* (9.8% — under convex rows, random features are informationless; all accuracy is earned feature learning); solving *during* training collapsed the search — at a block optimum the selection landscape flattens (envelope theorem): *a perfect readout is a bad teacher*.

MCTS over mutants (adapting Remy’s **The_Monte** engine): edit-level trees fail structurally — without re-rooting, progressive widening spends any budget within ~ 10 edits of init (chance); with commits, lookahead costs $5\times$ throughput for one level of foresight; a tuned cold-UCB tree reaches depth 123 and 39% val but stays $\sim 10\times$ behind flat annealing per FLOP. **Macro-MCTS** fixes both failure modes: each node is a *chunk of delta-annealing under a strategy* (operator mix $\times$ step scale) — the tree plans the schedule, not the synapses. Result: 87.1% at matched FLOPs, parity with the hand-tuned annealer, **with the curriculum discovered by search**: coarse blobs alternating functional-graph moves early, inhibition mid-run, ultra-fine transfers last — independently rediscovering the hand-designed schedule and the bandit’s late-inhibition insight. Scaled (10 macros, $3\times$ budget): 89.6%.

7 Quantization: the finite-lattice study

Fixing the update quantum makes weight space finite — but $\delta=1/32$ yields $\sim 10^{81,211}$ networks (naming one takes only ~ 270 kbits: the honest size of the learning problem). STE gradients vs a lattice-native annealer (exact δ -token transfers):

δ	levels	STE backprop	lattice annealer
—	∞	97.2%	87.4%
0.02	51	96.9%	90.0%
0.05	21	96.8%	90.8%
0.1	11	95.4%	88.8%
0.2	6	94.3%	84.9%
0.5	3	84.5%	38.4%

No crossover — but quantization *helps* gradient-free search absolutely (21 levels: best flat-annealing result of the day, three moves only), and lattice-native training crushes post-hoc rounding (84.9% vs 66.3% at $\delta=0.2$).

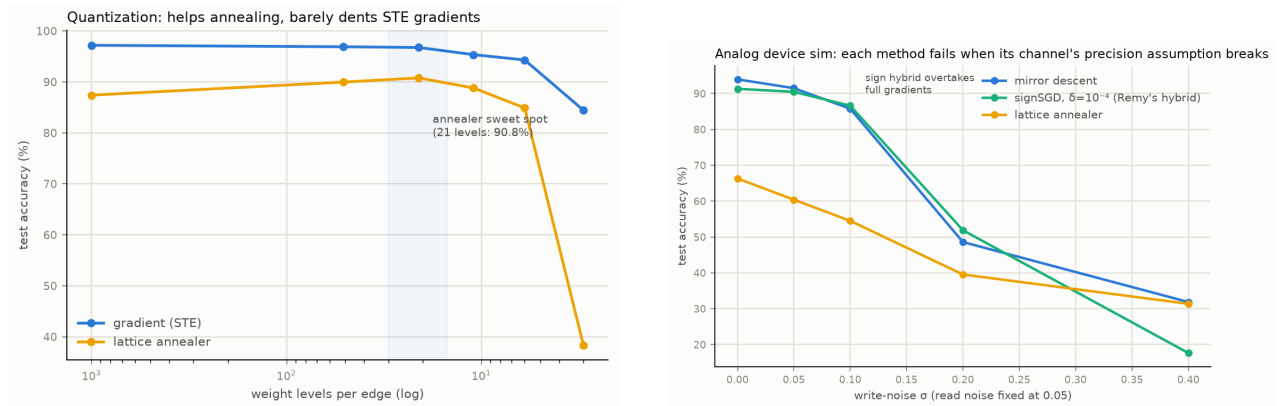


Figure 2: Left: quantization helps annealing, barely dents STE. Right: analog noise — each method fails when its channel’s precision assumption breaks; the fixed- δ sign hybrid overtakes full gradients at moderate write noise.

8 The analog-noise regime

Simulated device physics: write noise ($w \leftarrow w e^{\sigma \xi}$ on every *written* weight — you pay only for rows you write) and read noise ($z \pm \sigma \xi$ per training forward). At read $\sigma=0.05$:

write σ	mirror descent	signSGD $\delta=10^{-4}$	lattice annealer
0	93.9	91.3	66.3
0.05	91.5	90.5	60.4
0.1	85.8	86.6	54.5
0.2	48.6	51.9	39.5
0.4	31.9	17.7	31.4

The symmetric law: **gradients are robust to read noise, fragile to write noise; selection is the reverse**. The sign hybrid with δ matched to the write-noise floor overtakes full mirror descent at moderate noise: *the update quantum must exceed the write-noise floor — precision below the floor is worse than a blunt step*.

9 Interpretability

The geometry’s clearest win: every row is a budgeted convex combination minus bounded inhibition, so magnitudes have fixed units end-to-end and composed products $W_1 W_0$ are directly comparable digit templates. Interactive artifacts: topology diagram and 3D network explorer (three models, top- k slider, hover cones, click-to-inspect weight images and composed receptive fields).

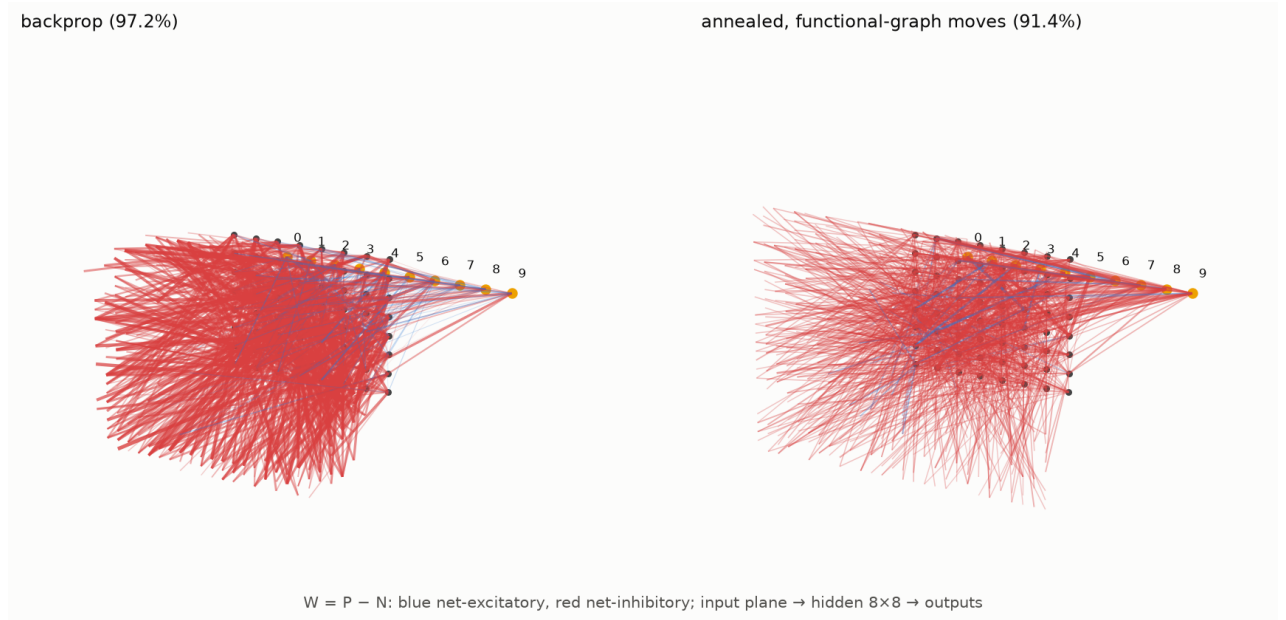


Figure 3: $W = P - N$ in 3D: blue net-excitatory, red net-inhibitory, saturation $\propto |W|$. Backprop wires sparsely and surgically; the annealed functional-graph model suppresses whole hidden communities.

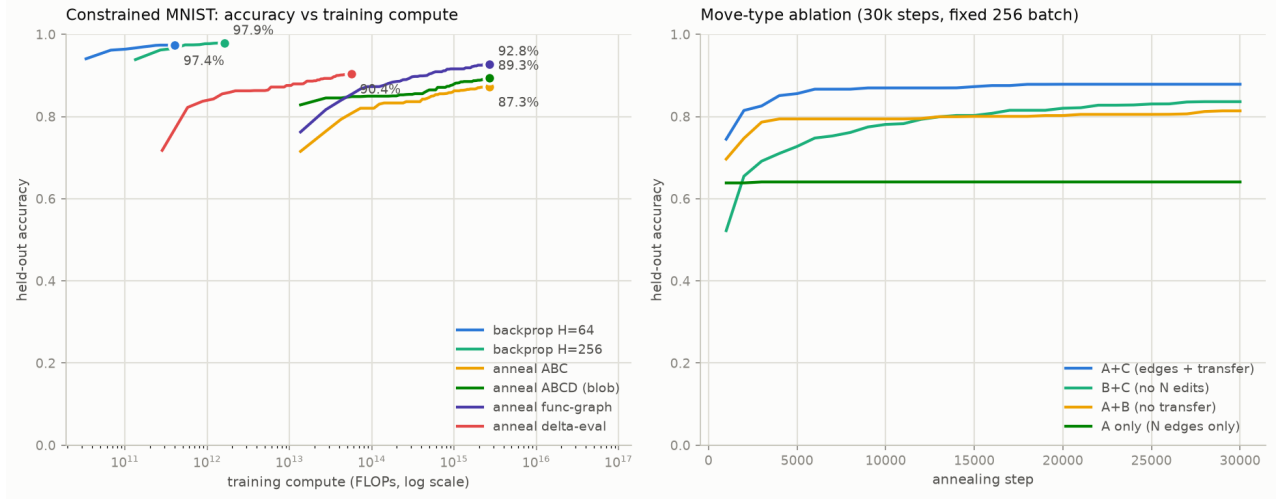


Figure 4: Accuracy vs training compute (left) and move-type ablations (right).

10 Final standings and thesis

Method	Test	FLOPs	Wall
Unconstrained MLP (H=128, Adam)	98.2%	4×10^{11}	2 s
Backprop + reparam (H=256)	97.7%	1.6×10^{12}	8 s
Backprop + reparam (H=64)	97.2%	4×10^{11}	5 s
Mirror descent (native simplex)	95.0%	1.7×10^{12}	18 s
signSGD fixed- δ	92.7%	8×10^{11}	6 s
Anneal + functional graph	91.4%	2.7×10^{15}	562 s
Lattice annealer $\delta=0.05$	90.8%	1.0×10^{13}	187 s
Macro-MCTS (10 strategies)	89.6%	3.5×10^{13}	770 s
Composite AGGCEH (delta eval)	89.4%	1.5×10^{13}	227 s
Edit-level MCTS (best)	~39% val	1.7×10^{12}	—

Thesis. Optimization methods are ranked by the information rate and precision of the channel they consume — full gradient magnitudes > gradient signs > candidate rankings — and the winner at any operating point is the method whose channel assumptions the substrate can still honor. Clean digital hardware honors Adam’s; quantized and write-noisy substrates progressively hand the advantage to sign-steps and selection. Structural priors are worth bits exactly when the search signal is information-poor, and near nothing once gradients flow. Deliberation pays at the strategy level, never at the synapse level.